SECTION 1: POWER INPUT, FILTERING & CURRENT SENSING

VBAT

GND

SHUNT_NEG

SECTION 2: MICROCONTROLLER & PERIPHERALS (STM32G071)

+3V3

CURR_SENSE

VBAT_SENSE

PWM_IN

TELEMETRY

STATUS_LED

SWDIO

SWCLK

NRST

BEMF_A

BEMF_B

BEMF_C

BEMF_NEUTRAL

SECTION 3: 3-PHASE GATE DRIVER (FD6288Q)

VCC_DRV

HIN_A

LIN_A

HIN_B

LIN_B

HIN_C

LIN_C

BOOT_A

BOOT_B

BOOT_C

GATE_HA

GATE_LA

GATE_HB

GATE_LB

GATE_HC

GATE_LC

SECTION 4: 3-PHASE INVERTER BRIDGE & BEMF SENSING

MOT_A

MOT_B

MOT_C

Unrouted Base Design for KiCad & FreeRouting
Input: 3S-6S LiPo (9V - 25.2V) | Continuous 35A / Peak 45A
MCU: STM32G071KBT6 / Driver: FD6288Q / 6x N-FETs 40V 60A
Autonomous Drone Systems

Sheet: /
File: custom_esc_35a.kicad_sch

Title: CUSTOM 3S-6S 35A/45A SINGLE BLDC ESC

Size: A3	Date: 2026-11-11	**Rev: v1.0**
KiCad E.D.A. 10.0.0		Id: 1/1